SOHAM KAMAT

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EDUCATION

The University of Washington, Seattle, WA 2025 - 2028 (Expected)
B.S. Mechanical Engineering
Del Norte High School, San Diego, CA 2021 - 2025

Skills:

Design & CAD: SolidWorks (CSWA & CSWP Certified), Siemens NX, Autodesk Inventor, Fusion 360, OnShape, Blender
Fabrication: Manual Milling, Welding, 3D Printing, Composite Layup (Carbon Fiber & Fiberglass)
Programming & Tools: Python, C/C++, Java, JavaScript, HTML, Bash, PowerShell, MATLAB

EXPERIENCE

UW Steel Bridge Team, Seattle, WA Oct 2025 - Present
- Placed 2nd overall at the regional competition (University of Alaska Anchorage), qualifying for the AISC Student Steel Bridge National Finals, where the team placed 18th nationally.
- Fabricated and machined steel structural members to exact specifications for the AISC Steel Bridge Competition.
- Assembled and test-fit competition bridge with a construction subteam, reworking components to resolve fit-up discrepancies against design drawings.

UW Society for Advanced Rocket Propulsion (SARP) Team, Seattle, WA Oct 2025 - Present
- Machined structural and propulsion components on a manual mill and lathe from engineering drawings, verifying dimensional tolerances against specifications before assembly.
- Researched 3D-printed tooling methods for composite structural couplers, developing carbon fiber and fiberglass layup processes across mold geometry, surface preparation, and release agents.
- Prepared and cut aluminum and steel stock into flight-ready rocket hardware for propulsion systems.

CyberPatriot (Cybersecurity Competition), San Diego, CA Jul 2021 - Feb 2025
- Secured Windows and Linux systems against live exploits in timed rounds, competing nationally to achieve the fastest and most thorough system remediation.
- Identified and patched vulnerabilities across user accounts, services, and firewall rules; configured group policy and administrative controls.

San Diego Supercomputer Center Internship, San Diego, CA Jun - Aug 2024
- Built cyberinfrastructure expanding public access to AI tools through hosted Jupyter notebooks.
- Maintained a Python, YAML, and Jekyll-based deployment pipeline for website metadata and content updates.

PROJECTS

Phone Holder – Mechanical Design SolidWorks, 3D Printing
- Designed a multi-part moving assembly with a cutout engineered to fit an Anker MagGo power bank; modeled, assembled, and motion-tested each component in SolidWorks before 3D printing.

Sim Racing Wheel – Mechanical & Electronics Design SolidWorks, Arduino, 3D Printing
- Designed a 3D-printed sim racing wheel using a magnetic rotary encoder and a microcontroller to report steering position over USB across roughly 540° of rotation.

LEADERSHIP & COMMUNITY INVOLVEMENT

Akshaya Patra Foundation, San Diego Chapter – Co-Head 2021 - 2025
- Managed 70 volunteers and coordinated regional fundraising efforts, raising over $120,000 for school meal programs in India.

Rural Education Advancement Trust – Coordinator 2022 - 2025
- Created STEM content reaching 4,000+ rural students and organized virtual learning events and competitions.

HONORS AND AWARDS

- President's Volunteer Service Award — National Level (2024)
- National Merit Commended Student (2023)